MEETING ROOM BOOKING SYSTEM

User Acceptance Test (UAT) & Readiness Assessment

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Document Story

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User Acceptance Test (UAT) & Readiness Assessment V.0.1	Initial Draft UAT Specification	8 Feb 2026	TNR,TNP, JTS	Draft
User Acceptance Test (UAT) & Readiness Assessment_V.0.2	Update Test Cases based on SRS V.1.0	12 Feb 2026	SANTACLAUS TEAM	Reviewed
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Acronym
TNR = Thanarak Chumsawat(Product Owner)
TNP = Tanapat Chaitong (Scrum Master)
PTP = Pathompon Leejang (Developer)
PST = Pasit Petoumphun (Developer)
JTS = Wijitsin Fangyen (Developer)

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Chapter One | Introduction

1.1 Purpose

The purpose of this document is to outline the User Acceptance Testing (UAT) for the "Meeting Room Booking System". It serves as a verification tool to ensure that the developed system meets the business requirements and functional specifications defined in the Software Requirement Specification (SRS). The primary goal is to confirm the system's readiness for deployment.

1.2 Scope of Testing

The testing scope encompasses all functional modules described in the Product Backlog:

1.2.1 **Authentication:** Login and Role-based access control (User vs. Admin).
1.2.2 **Booking Management:** Real-time availability, booking creation, and conflict detection.
1.2.3 **Process Automation:** Email notifications, Check-in validation, and Auto-release logic.
1.2.4 **dmin & Management:** Room management, VIP approval workflow, and facility reporting.

1.3 Definitions & Acronyms

- **SRS:** Software Requirement Specification.
- **Double Booking:** A critical failure where two users book the same room at the same time.
- **Ghost Booking:** A booking where the user fails to show up.
- **Race Condition:** A system behavior where the output depends on the sequence or timing of other uncontrollable events.

Chapter Two | Test Environment

The UAT was conducted in a production-like environment using the following configurations

2.1 Hardware Environment

- **Development/Client Workstations:**
    - Apple MacBook Pro M4
    - HP Victus 16 (s0095ax)
    - Acer Nitro V 15 (ANV15-51-50UV)
- **Mobile Devices:** iPhone 17 (iOS 18), Samsung Galaxy S24 (Android 15)

2.2 Product Features

- **Cloud Infrastructure:** AWS EC2 (Amazon Elastic Compute Cloud)
- **Database:** Supabase (PostgreSQL)
- **Backend Technology:** Dockerized Application (Antigravity Framework)

Chapter Three | Acceptance Criteria & Methodology

3.1 Status Criteria

Each test case is evaluated based on the following criteria:

Status	Color Code	Definition	Decision Criteria	Action Required
Ready / Passed	**[Green]**	**Passed / Ready for Production**	• Functionality meets all SRS requirements (100%). • No bugs, errors, or defects found. • Performance metrics are met (e.g., load time < 2 seconds).	• **Approved for Go-Live.** • Project Sign-off can proceed.
Warning / Partial	**[Yellow]**	**Conditional Pass / Monitor**	• Core functionality works as intended. • Minor defects found (e.g., UI glitches, typos) that do not block critical operations. • Slight performance latency observed.	• **Approved for Go-Live with Caution.** • Fixes must be scheduled for the immediate next patch. • Workarounds must be documented for users.
Critical / Failed	**[Red]**	**Failed / Critical Block**	• Core functionality failed (System Crash, Error 500). • Data integrity issues detected (e.g., **Double Booking** is possible). • Critical features are incomplete or missing.	• **STOP Go-Live.** • Immediate fixes required. • Full re-test is mandatory until the status becomes Green.

- **[Pass] Green:** The function works exactly as specified in the SRS. No errors observed.
- **[Warning] Yellow:** The function works but has minor issues (e.g., UI glitches, slight latency) that do not stop operation.
- **[Fail] Red:** The function fails to execute, produces incorrect results, or causes a system crash. **Critical functionality failures must be fixed before Go-Live.**

Chapter Four | UAT & Readiness Assessment Matrix

4.1 Module: Authentication System

4.1.1 Reference: UC-01, AD-01

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
AUTH-01	UC-01	**User Login (Success)**	Valid User Account exists.	1. Enter valid User credentials. 2. Click Login.	System redirects to **User Dashboard**.	As expected.	**[Pass]**	Pathompon
AUTH-02	UC-01	**Admin Login (Success)**	Valid Admin Account exists.	1. Enter valid Admin credentials. 2. Click Login.	System redirects to **Admin Console**.	As expected.	**[Pass]**	Pathompon
AUTH-03	UC-01	**Invalid Login**	None.	1. Enter incorrect password. 2. Click Login.	System displays "Invalid username or password".	As expected.	**[Pass]**	Pathompon
AUTH-04	N/A	**Logout**	User is logged in.	1. Click "Sign Out".	System redirects to Login Page and clears session.	As expected.	**[Pass]**	Pathompon

4.2 Module: Booking System (Core Functionality)

4.2.1 Reference: Feature #1 Booking System

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
BOOK-01	SRS-003	**Dashboard Performance**	Logged in.	1. Load the dashboard. 2. Measure load time.	Calendar loads within **2 seconds**.	Load time is approx 2.5s.	**[Warning]**	Pasit
BOOK-02	SRS-005	**Create Booking (Happy Path)**	Room is Available.	1. Select Date/Time. 2. Fill Topic/Attendees. 3. Click Confirm.	Booking saved. Status becomes **Red**.	As expected.	**[Pass]**	Wijitsin
BOOK-03	SRS-007	**Past Time Validation**	None.	1. Select yesterday's date. 2. Attempt to book.	System displays error: "Cannot book past time".	As expected.	**[Pass]**	Pathompon
BOOK-04	SRS-009	**Zero Double Booking (Logic)**	Room already booked by User A.	1. User B tries to book same slot. 2. Click Confirm.	System displays error: "Room is not available".	As expected.	**[Pass]**	Pathompon
BOOK-05	SRS-011	**Concurrency Stress Test**	Slot is free.	1. User A & B click Confirm **simultaneously**(Race Condition).	Only one user succeeds. The other gets an error.	**BOTH users succeeded.**(Double Booking occurred).	**[Fail]**	Pathompon

4.3 Module: Cancellation System

Reference: URS-04, SRS-012

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
CNCL-01	SRS-012	**Cancel Own Booking**	Active booking exists.	1. Navigate to My Bookings. 2. Click Cancel.	Status changes to "Cancelled". Room becomes Green.	As expected.	**[Pass]**	Pasit
CNCL-02	SRS-013	**Cancel Others' Booking**	Logged in as User A.	1. Try to access User B's booking ID via URL.	Access Denied or Cancel button hidden.	Button is hidden. Secure.	**[Pass]**	Pasit

4.4 Module: Notification & Process Automation

Reference: Feature #2 Notification & Check-in

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
AUTO-01	SRS-015	**Email Confirmation**	Booking created.	1. Check User Inbox.	Email received with correct Topic/Time/Room.	As expected.	**[Pass]**	Wijitsin
AUTO-02	SRS-018	**Check-in Window (Before)**	Meeting starts in 30 mins.	1. View Booking Details. 2. Look for Check-in button.	Button is **NOT visible**.	As expected.	**[Pass]**	Wijitsin
AUTO-03	SRS-018	**Check-in Window (During)**	Meeting starts in 5 mins.	1. View Booking Details. 2. Look for Check-in button.	Button is **VISIBLE**.	Visible but requires page refresh.	**[Warning]**	Wijitsin
AUTO-04	SRS-019	**Ghost Booking Auto-Release**	User did **not**check in within 15 mins.	1. Wait 16 mins after start time. 2. Check Room Status.	System cancels booking. Room becomes Green.	**Room remains Red.**(Cron job failure).	**[Fail]**	Pathompon

4.5 Module: Admin & Management

Reference: Feature #4 Admin Management

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
ADM-01	SRS-021	**Add New Room**	Admin logged in.	1. Add "Jupiter Room". 2. Check User Dashboard.	"Jupiter Room" appears immediately.	As expected.	**[Pass]**	Tanapat
ADM-02	SRS-022	**Disable Room**	Room is Active.	1. Set status to "Maintenance". 2. User tries to book.	Room is greyed out/unselectable.	As expected.	**[Pass]**	Tanapat
ADM-03	SRS-024	**VIP Booking Request**	User books VIP Room.	1. User submits booking.	Status is "Pending Approval".	As expected.	**[Pass]**	Thanarak
ADM-04	SRS-027	**Approve VIP Request**	Request is Pending.	1. Admin clicks Approve.	Status becomes "Confirmed". User gets email.	As expected.	**[Pass]**	Thanarak

4.6 Module: Facility Reporting

Reference: URS-10

ID	SRS Ref	Test Case Name	Pre-condition	Test Steps	Expected Result	Actual Result	Status	Owner
RPT-01	SRS-028	**Report Issue**	Logged in.	1. Select Room. 2. Select "Projector Broken". 3. Submit.	Report saved to database.	As expected.	**[Pass]**	Pasit
RPT-02	SRS-029	**Admin Notification**	Report submitted.	1. Check Admin Dashboard.	Notification appears immediately.	Notification delayed by 5 mins.	**[Warning]**	Pasit

4.7 System Readiness Assessment Summary

This section summarizes the overall system readiness based on the critical path items tested in sections 4.1 to 4.6. It serves as a high-level scorecard for the "Go-Live" decision.

Critical Path Item	Success Criteria	Workaround/ Contingency Plan	Planned Start Date	Planned End Date	Responsible Party	Previous Status	Current Status	Status Criteria	Comments/ Corrective Action
1-Critical Technical Issues						**[RED]**	**[RED]**	1+ Red = Red	**CRITICAL BLOCKER**
1.1 Zero Double Booking	System must prevent simultaneous bookings (Race Condition) 100%.	Admin manually checks logs and cancels duplicates.	10 Jan 2026	14 Feb 2026	Pathompon	**[YELLOW]**	**[RED]**	Yellow = 1 medium outstanding	**FAILED (Ref: BOOK-05):**Race Condition detected. Database Lock required.
1.2 Auto-Release (Ghost Booking)	System must auto-cancel if no Check-in within 15 mins.	Admin monitors dashboard and resets room status manually.	12 Jan 2026	14 Feb 2026	Pathompon	**[GREEN]**	**[RED]**	See Above	**FAILED (Ref: AUTO-04):**Cron Job not triggering.
1.3 Dashboard Performance	Dashboard must load availability within 2 seconds.	User manually refreshes the browser.	10 Jan 2026	14 Feb 2026	Pasit	**[YELLOW]**	**[YELLOW]**	See Above	**WARNING (Ref: BOOK-01):**Avg load time ~2.5s.
2-Functional Readiness						**[GREEN]**	**[GREEN]**		**PASS**
2.1 Authentication Module	User/Admin can login/logout correctly.	IT Support creates temporary accounts.	08 Jan 2026	12 Feb 2026	Pathompon	**[GREEN]**	**[GREEN]**	See Above	Passed all cases. Stable.
2.2 Booking Flow (Standard)	User can book a room in < 4 clicks.	Admin books via backend.	10 Jan 2026	12 Feb 2026	Wijitsin	**[GREEN]**	**[GREEN]**	See Above	Passed Happy Path.
2.3 Admin Management	Admin can manage rooms/approve VIP.	Manual approval via email/phone.	12 Jan 2026	12 Feb 2026	Thanarak	**[GREEN]**	**[GREEN]**	See Above	Passed all admin functions.
3-Go-Live Decision						**[RED]**	**[RED]**		**STOP Go-Live**

Chapter 5: Defect Log & Corrective Actions

Based on the UAT execution, the following critical defects were identified and require immediate attention:

5.1 [Critical] Race Condition (BOOK-05):

- **Issue:** Two users can book the same slot if requests are sent at the exact same millisecond.
- **Action:** Implement Database Transaction Isolation (Serializable) or Row Locking in Supabase.
- **Responsible:** Pathompon (Developer).

5.2 [Critical] Auto-Release Failure (AUTO-04):

- **Issue:** Rooms are not released when users fail to check in. The server-side scheduler (Cron Job) is not triggering.
- **Action:** Debug the AWS Lambda/Cron configuration to ensure the script runs every minute.
- **Responsible:** Pathompon (Developer).

5.3 [Warning] Dashboard Latency (BOOK-01):

- **Issue:** Loading times exceed the 2-second KPI (approx. 2.5s).
- **Action:** Optimize SQL queries and add indexes to the `bookings` table.
- **Responsible:** Pasit (Developer).

Chapter 6: Conclusion & Sign-off

6.1 Summary of Results

- **Total Test Cases:** 18
- **Passed ([Green]):** 13
- **Warning ([Yellow]):** 3
- **Failed ([Red]):** 2

6.2 Conclusion

The system Core Functionality (Booking, Admin, Auth) is stable. However, the **Critical Issues regarding Concurrency (Double Booking) and Auto-Release** must be resolved before the system can be declared "Ready for Production".

6.3 Sign-off

We certify that the UAT has been performed and the results recorded above are accurate.

Product Owner: Thanarak (Mr.ThanarakChumsawat)

Date: 11 Feb. 2026